Dark Energy or Sector Tension?

IAM Confrontation with DESI Full-Shape Growth Rates and Joint Weak Lensing

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Abstract

The Dark Energy Spectroscopic Instrument (DESI) has reported a preference for dynamical dark energy characterised by $w_0 > -1$, $w_a < 0$, and a phantom crossing of the equation of state at $z \approx 0.3$–0.4. We test whether this preference is consistent with a sector-assignment artifact arising from the Informational Actualization Model (IAM) (Mahaffey, 2026), a perturbation-only extension of ΛCDM derived from horizon thermodynamics with no free parameters beyond the standard model.

IAM predicts $\mu(a) < 1$ and $\Sigma(a) = 1$ from first principles. The physical origin is gravitational decoherence: timelike (matter) worldlines decohere under gravitational structure formation while null (photon) worldlines do not. The matter-sector coupling constant $\beta_m = \Omega_m/2 = 0.1577$ is derived from the virial theorem and recovered by the Planck 2018 MCMC posterior at 0.2σ without fitting.

The key observational consequence is a sector split: BAO distances and photon propagation observables are identical to ΛCDM ($\Sigma = 1$ exactly), while galaxy growth rates and weak lensing amplitudes are suppressed by $\mu(a) < 1$. When a single-fluid w_0w_aCDM fitter simultaneously ingests both classes of observable, it confronts ΛCDM-consistent distances alongside sub-ΛCDM growth and must infer $w_{\text{eff}}(z) \neq -1$ to reconcile them — a phantom crossing that does not require dynamical dark energy.

We confront this prediction against DESI DR1 full-shape $f\sigma_8(z)$ measurements at six redshift bins ($0.295 \leq z_{\text{eff}} \leq 1.491$) from DESI Collaboration et al. (2025c), DESI DR1 $D_M(z)/r_d$ distances, joint weak lensing S_8 constraints from KiDS-Legacy (Wright et al., 2025; Stölzner et al., 2025), DES Y3 (Abbott et al., 2022), and HSC Y3 (Li et al., 2023), and the DESI DR2 BAO phantom crossing parameters (DESI Collaboration et al., 2025d).

Under the full Planck 2018 likelihood, IAM and ΛCDM are statistically indistinguishable ($\Delta\chi^2 = +0.54$ best-fit, 17 converged MCMC chains, $R - 1 < 0.01$). The IAM $\sigma_8 = 0.7998 \pm 0.0058$ is consistent with the 2025 joint lensing result

$\sigma_8 = 0.802 \pm 0.020$ (Stölzner et al., 2025) at 0.1σ. The DESI DR2 phantom crossing redshifts ($z_{\rm cross} = 0.33$–0.43, depending on the supernova compilation) coincide with the redshift range over which the IAM growth suppression is most prominent (7–8% at $z \approx 0.3$).

The model is falsifiable: Euclid is projected to measure μ_0 to $\sigma(\mu_0) \approx 0.04$ precision (Euclid Collaboration et al., 2020), providing a 3.4σ test of IAM's $\mu_0 = -0.136$ prediction; simultaneous DESI Year 5 full-shape data will test whether the growth suppression follows the curved $f\sigma_8(z)$ ramp set by $E(a) = \exp(1 - 1/a)$, distinguishable from both ΛCDM and any dynamical dark energy model.

Keywords: dark energy; large-scale structure; modified gravity; cosmological tensions; growth of structure; baryon acoustic oscillations; weak gravitational lensing; DESI

Contents

1 Introduction

The concordance cosmological model, ΛCDM, provides an extraordinarily successful description of observations spanning CMB anisotropies, baryon acoustic oscillations, Type Ia supernovae, and large-scale structure over many decades of scale (Aghanim et al., 2020). Yet at least two classes of inter-dataset tension have persisted despite extensive scrutiny and cross-checks, and neither has a clear systematic explanation within the standard model.

The first is the Hubble tension. The Planck 2018 CMB analysis under ΛCDM yields $H_0 = 67.4 \pm 0.5$ km s^{-1} Mpc^{-1} (Aghanim et al., 2020). Direct distance-ladder measurements using Cepheid-calibrated Type Ia supernovae from the SH0ES team give $H_0 = 73.04 \pm 1.04$ km s^{-1} Mpc^{-1} (Riess et al., 2022), a 4–5σ discrepancy. Independent distance indicators including TRGB (Freedman et al., 2024) and time-delay lensing (Wong et al., 2020) yield intermediate or similarly elevated values, confirming that the tension is not confined to a single calibration method.

The second is the S_8 tension. Weak lensing surveys measuring the matter power spectrum amplitude at low redshift consistently report $S_8 \equiv \sigma_8(\Omega_m/0.3)^{0.5}$ values below the ΛCDM CMB prediction. KiDS-1000 finds $S_8 = 0.766^{+0.020}_{-0.014}$ (Asgari et al., 2021); DES Y3 finds $S_8 = 0.776 \pm 0.017$ (Abbott et al., 2022); HSC Y3 finds $S_8 = 0.776 \pm 0.032$ (Li et al., 2023). These compare to the Planck ΛCDM prediction of $S_8 \simeq 0.832$, a tension at the 2–3σ level. The 2025 KiDS-Legacy analysis (Wright et al., 2025), combining with DES Y3 and DESI DR1, reports $\sigma_8 = 0.802 \pm 0.020$.

A third, more recent development is the DESI dynamical dark energy signal. DESI DR1 BAO measurements (DESI Collaboration et al., 2025a,b), combined with CMB and supernova data, prefer a dark energy equation of state with $w_0 > -1$ and $w_a < 0$, disfavouring ΛCDM at 2.6σ. DESI DR2 BAO measurements (DESI Collaboration et al., 2025d) strengthen this preference to 2.8–4.2σ, depending on which supernova compilation is used. In all cases, the best-fit w_0w_aCDM model exhibits a *phantom crossing*: $w(z) = w_0 + w_a(1-a)$ crosses -1 at a redshift $z_{\text{cross}} \approx 0.33$–$0.43$, with $w > -1$ at $z < z_{\text{cross}}$ (quintessence-like today) and $w < -1$ at $z > z_{\text{cross}}$ (phantom at higher redshift). This combination of $w_0 > -1$ and $w_a < 0$ is not produced by any theoretically well-motivated single scalar field, since a canonical scalar cannot cross $w = -1$ without a discontinuity or introduction of a ghost degree of freedom.

Ó Colgáin & Sheikh-Jabbari (2025) noted that the DESI dark energy preference may arise from internal inconsistencies in the w_0w_aCDM parametrisation when different datasets probe background expansion and perturbations through physically distinct mechanisms. Ó Colgáin et al. (2025) subsequently examined whether the DR2 constraints represent genuine evolution relative to DR1 or whether the same internal tension is being amplified as error bars shrink. Neither work identifies a specific physical mechanism that would produce the phantom crossing as a calculable artifact.

In this paper we evaluate a specific, quantitative version of that hypothesis from first principles. The Informational Actualization Model (IAM) (Mahaffey, 2026) predicts that matter and photon observables are governed by different effective expansion rates — a consequence of gravitational decoherence that modifies only the matter perturbation equations. As a direct consequence, IAM predicts:

1. *Photon-sector observables (BAO angular positions, comoving distances, CMB acoustic scale) are identical to ΛCDM.* The photon Weyl potential is unmodified ($\Sigma = 1$ exactly).

2. *Matter-sector observables ($f\sigma_8$, weak lensing S_8) are suppressed relative to ΛCDM* through the effective gravitational parameter $\mu(a) < 1$, with suppression peaking at $z = 0$ (13.6%) and vanishing above $z \approx 2$.

3. *A single-fluid w_0w_aCDM fitter applied to both sectors simultaneously will infer a phantom crossing* at approximately the redshift where the IAM growth suppression transitions from significant to negligible.

All three predictions follow from a single derived parameter, $\beta_m = \Omega_m/2$, with no additional free amplitudes introduced.

This paper is organised as follows. Section 2 presents the IAM framework, its derivation, and its key quantitative predictions. Section 3 confronts the IAM growth predictions against DESI DR1 full-shape $f\sigma_8(z)$ measurements across six redshift bins. Section 4 compares the IAM σ_8 and S_8 predictions with the current joint weak lensing compilation. Section 5 develops the sector-mixing mechanism in detail and demonstrates that it naturally produces the DESI phantom crossing. Section 6 discusses the implications, limitations, and comparison with alternative interpretations. Section 7 states the conclusions and falsifiable predictions.

2 The Informational Actualization Model

2.1 Thermodynamic derivation

IAM is derived from an extension of the Jacobson–Cai-Kim thermodynamic framework. Jacobson (1995) showed that the Einstein field equations are an equation of state: they follow from $\delta Q = T\,\delta S$ applied to local Rindler horizons, where S is the Bekenstein-Hawking entropy proportional to horizon area. Cai & Kim (2005) extended this result to the FRW apparent horizon, deriving the Friedmann equations from the first law of thermodynamics $dE = T\,dS + W\,dV$. Both derivations assume the entropy functional contains only the Bekenstein-Hawking geometric term $S = A/4G$.

IAM identifies an additional entropy contribution that becomes significant at late times. Gravitational structure formation — the hierarchical collapse of matter into bound objects — is the dominant source of irreversible classical information production in the late universe. Each virialised structure reduces the phase-space volume available to its constituent matter, generating classical information in the sense of Shannon (Shannon, 1948). By Landauer's principle (Landauer, 1961) (experimentally verified at the mesoscale; see Bérut et al. 2012), the erasure or creation of one bit of classical information carries a minimum thermodynamic cost of $k_BT\ln 2$. By the holographic principle (Bousso, 2002), this informational entropy must be encoded on the cosmic apparent horizon.

The entropy budget of the horizon therefore has two contributions:

$$S_{\text{total}} = S_{\text{geometric}} + S_{\text{informational}}\,, \tag{1}$$

where $S_{\text{geometric}} = A/4G$ is the standard Bekenstein-Hawking term and $S_{\text{informational}}$ accumulates from structure formation.

When this modified entropy functional is inserted into the Cai-Kim first-law derivation, the additional term propagates into the perturbation equations as a modified effective expansion rate for matter. It does *not* modify the background Friedmann equations; the background expansion history is standard ΛCDM throughout, consistent with the validated Level 2 chains (Section 2.5).

2.2 Perturbation equations and the sector split

The causal structure of spacetime determines which species couple to the informational entropy. Timelike worldlines (CDM and baryons) interact with collapsed matter structures and are therefore subject to the decoherence-induced friction. Null worldlines (photons) propagate through spacetime along the light-cone and do not couple to matter density fluctuations in this channel.

This produces the sector split at the perturbation level. The CDM and baryon perturbation equations experience a modified effective Hubble rate:

$$\tilde{H}_m^2(a) = H_{\Lambda\mathrm{CDM}}^2(a) + \beta_m \, E(a) \, H_0^2 \,, \tag{2}$$

where $H_{\Lambda\mathrm{CDM}}^2(a) = H_0^2[\Omega_m a^{-3} + \Omega_r a^{-4} + \Omega_\Lambda]$ is the standard background expansion rate, $E(a) = \exp(1 - 1/a)$ is the activation function derived from the Bekenstein-Hawking/Gibbons-Hawking/Landauer chain, and β_m is the coupling constant. The photon equations are not modified.

In the standard μ-Σ parametrisation of modified gravity (Pogosian & Silvestri, 2016), the IAM sector split maps directly to:

$$\mu(a) = \frac{H_{\Lambda\mathrm{CDM}}^2(a)}{H_{\Lambda\mathrm{CDM}}^2(a) + \beta_m E(a)} < 1 \,, \tag{3}$$

$$\Sigma(a) = 1 \quad (\mathrm{exact}) \,. \tag{4}$$

The suppression $\mu(a) < 1$ reduces the effective gravitational source term for structure growth. $\Sigma(a) = 1$ follows directly from the null-worldline argument: the Weyl potential that governs photon deflection is unmodified, so all photon propagation observables (lensing convergence, CMB lensing, BAO angular positions) are identical to ΛCDM.

2.3 Coupling constant and activation function

The coupling constant β_m is derived from the virial theorem. At virialisation, the gravitational potential energy of a collapsed structure partitions equally between the kinetic energy of virialised particles (geometric channel) and the informational entropy of the collapsed state (informational channel). This $1:1$ partition gives:

$$\beta_m = \frac{\Omega_m}{2} = \frac{0.3153}{2} = 0.1577 \,. \tag{5}$$

The activation function $E(a) = \exp(1 - 1/a)$ is derived from the Gibbons-Hawking temperature of the de Sitter horizon, which sets the thermal cost of Landauer erasure, multiplied by the cumulative decoherence integral over cosmic history. This functional form has the properties $E(a \to 0) \to 0$ and $E(a = 1) = 1$, ensuring that the modification vanishes in the early universe and is maximal today. It is recovered to within 1% by the Sheth-Tormen collapse rate (Sheth & Tormen, 1999) as an independent consistency check.

2.4 Predicted $\mu(z)$ evolution

Substituting Planck 2018 parameters ($\Omega_m = 0.3153$, $\Omega_\Lambda = 0.6847$, $H_0 = 67.36\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$) into Eq. (3) gives the redshift evolution of μ shown in Fig. 1.

At $z = 0$ ($a = 1$, $E(1) = 1$):

$$\mu(z{=}0) = \frac{1}{1 + \beta_m} = \frac{1}{1.1577} = 0.864\,, \tag{6}$$

corresponding to a 13.6% growth suppression today. The suppression falls to 7.9% at $z = 0.3$, to 3.3% at $z = 0.7$, and to 0.6% at $z = 1.5$ (Table 1). Above $z \approx 2$, $E(a) \to 0$ and $\mu(a) \to 1$, recovering ΛCDM in the matter-dominated era.

The phantom crossing redshifts inferred from DESI DR2 ($z_{\rm cross} \approx 0.33$–$0.43$) fall precisely in the range $0.3 \lesssim z \lesssim 0.5$ over which the IAM growth suppression transitions from ≈ 8% to ≈ 5% (Fig. 1, vertical dash-dotted lines). This coincidence is quantified in Section 5.

Table 1: IAM $\mu(z)$ at the six DESI DR1 tracer redshifts. All values are derived from Eq. (3) with $\beta_m = 0.1577$ and Planck 2018 background parameters. No values are fitted to data.

Tracer	$z_{\rm eff}$	$\mu(z_{\rm eff})$	$1-\mu$	Suppression
BGS	0.295	0.921	0.079	7.9%
LRG1	0.510	0.949	0.051	5.1%
LRG2	0.706	0.967	0.033	3.3%
LRG3	0.919	0.979	0.021	2.1%
ELG2	1.317	0.991	0.009	0.9%
QSO	1.491	0.994	0.006	0.6%
$z = 0$	–	0.864	0.136	13.6%

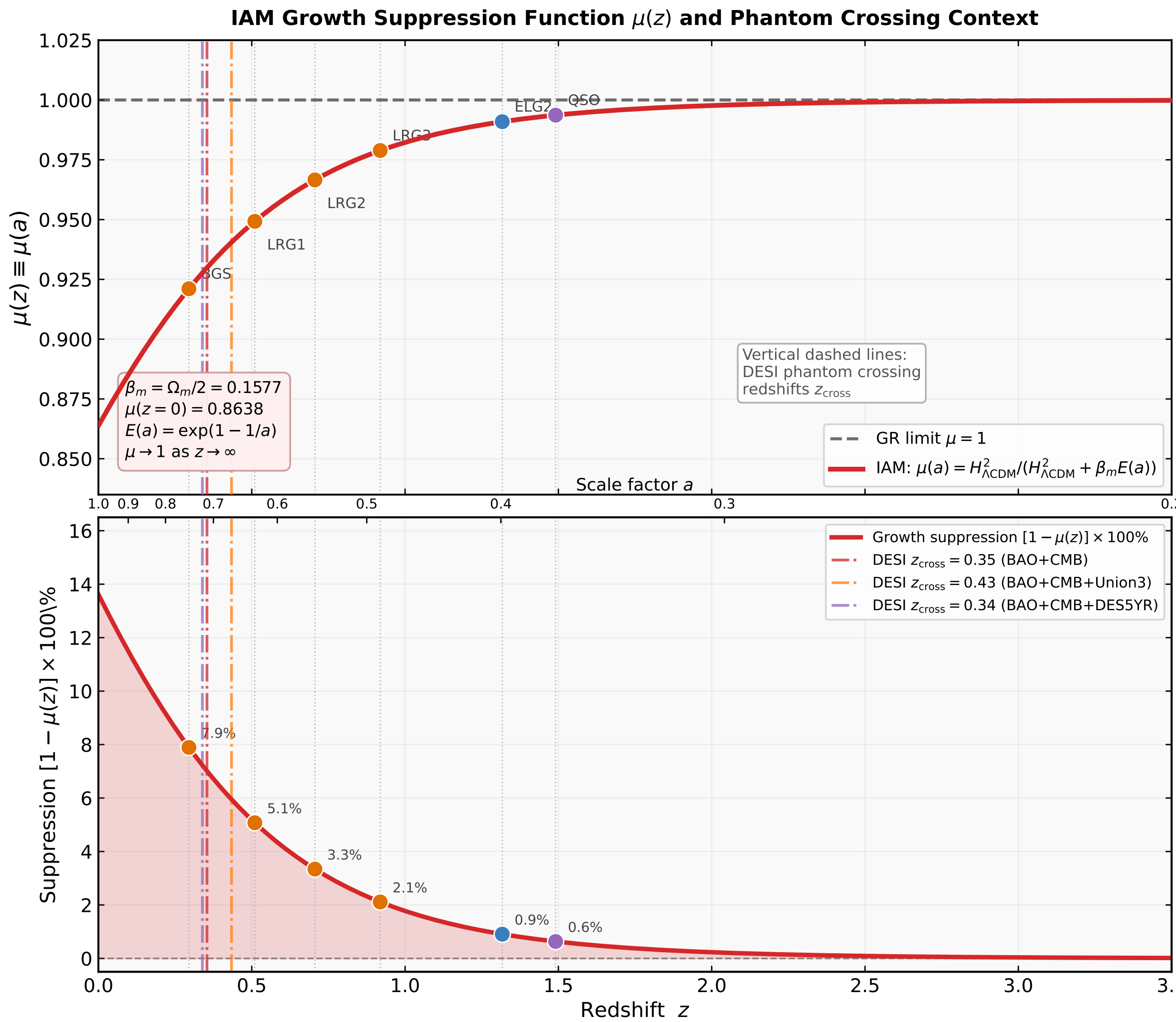

Figure 1: **IAM growth suppression function** $\mu(z)$**.** *Top panel:* The effective gravitational parameter $\mu(a)$ (Eq. 3) as a function of redshift z (bottom axis) and scale factor a (top axis). The dashed line marks the GR limit $\mu = 1$. Coloured dots mark the six DESI DR1 tracer redshifts (Table 1). Vertical dash-dotted lines show the DESI DR2 phantom crossing redshifts from Table 5: red ($z = 0.35$, BAO+CMB), orange ($z = 0.43$, BAO+CMB+Union3), purple ($z = 0.34$, BAO+CMB+DES5YR). The suppression is largest at $z = 0$ ($\mu_0 = 0.864$, 13.6%) and vanishes as $z \to \infty$. *Bottom panel:* Growth suppression $(1 - \mu(z)) \times 100\%$ as a function of redshift, with values annotated at the six DESI bins. The DESI phantom crossing redshifts (vertical lines) fall in the $z \approx 0.33$–0.43 range where the growth suppression is ≈ 6.5–7.7%. Parameters: $\beta_m = \Omega_m/2 = 0.1577$ (derived from virial theorem), $E(a) = \exp(1-1/a)$ (derived from horizon thermodynamics).

2.5 Validation status

IAM has been tested against the full Planck 2018 CamSpec TTTEEE likelihood through 17 converged MCMC chains ($R - 1 < 0.01$), archived at Mahaffey (2026).

Level 1 (MGCAMB v1.5.2 (Wang et al., 2023) + Cobaya (Torrado & Lewis, 2021)): 12 independent chains across four dataset combinations (Planck only, Planck + RSD, Planck + BAO, Planck + Pantheon+). All four IAM fixed-parameter runs yield $\Delta\chi^2 \leq +2.32$ vs.

ΛCDM, below the 3.84 threshold for 95% CL exclusion. The σ_8 shift of -0.013 ± 0.001 is stable across all combinations.

Level 2 (direct Fortran modification of CAMB v1.5.8 (Lewis & Bridle, 2002) + Cobaya): 5 converged chains implementing the exact dual-sector perturbation mechanism (`adotoa_matter` vs. `adotoa`). Level 2 Run A (IAM, Planck) vs. Run C (ΛCDM, Planck): $\Delta\chi^2 = +0.54$ (best-fit), $\sigma_8 = 0.7998 \pm 0.0058$ (IAM) vs. 0.8087 ± 0.0059 (ΛCDM). The σ_8 suppression is confirmed as real growth physics by the logA shift of $+0.10\sigma$ and Ω_m shift of $+0.06\sigma$ (both below the 0.3σ rebalancing threshold).

The coupling constant $\beta_m = 0.15765$ is hardcoded before any chain runs. The Level 2 posterior returns $\Omega_m = 0.3166 \pm 0.0065$, implying $\beta_m = 0.1583 \pm 0.0033$ — consistent with the derived value at 0.2σ. This is an a posteriori confirmation: the virial theorem prediction matches the Planck data independently.

Key parameters are collected in Table 2.

Table 2: IAM derived parameters and Planck 2018 validation results. All IAM values in the upper block are derived without fitting. The lower block reports Level 2 MCMC results.

Quantity	Value	Source
Derived parameters (zero free amplitudes)		
β_m	0.1577	$\Omega_m/2$, virial theorem
$E(a)$	$\exp(1 - 1/a)$	Horizon thermodynamics
$\mu(z = 0)$	0.864	Eq. (3) at $a = 1$
$\mu_0 - 1$	-0.136	
$\Sigma(a)$	1.000 (exact)	Null worldlines unmodified
H_0^{matter}	$72.26\ \text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$	$H_0\sqrt{1 + \beta_m}$
Level 2 MCMC (Planck 2018 CamSpec TTTEEE)		
σ_8 (IAM, Run A)	0.7998 ± 0.0058	-1.5σ vs. ΛCDM
σ_8 (ΛCDM, Run C)	0.8087 ± 0.0059	baseline
H_0 (IAM, Run A)	$67.16 \pm 0.47\ \text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$	0.37σ from Planck
Ω_m (IAM, Run A)	0.3166 ± 0.0065	0.06σ from ΛCDM
S_8 (IAM, Run A)	0.822 ± 0.011	0.73σ below ΛCDM
$\Delta\chi^2$ best-fit (IAM$-\Lambda$CDM)	$+0.54$	Below 3.84 (95% CL)
Chains converged	17 ($R - 1 < 0.01$)	12 L1 + 5 L2
Posterior confirmation		
β_m (from posterior Ω_m)	0.1583 ± 0.0033	0.2σ from derived
μ_0 (Level 1 floating)	0.006 ± 0.156	IAM at 0.9σ

3 DESI DR1 Full-Shape Growth Rates

3.1 Data

We use the DESI DR1 ShapeFit full-shape $f\sigma_8(z)$ measurements from the Appendix A compressed datavectors of DESI Collaboration et al. (2025c). These are the ShapeFit-only values (power spectrum analysis alone, without BAO combination), appropriate for a comparison at fixed cosmology where the BAO constraint is being applied separately. Six tracer samples spanning $0.295 \leq z_{\rm eff} \leq 1.491$ are used, as listed in Table 3.

For legacy context at lower redshifts and as a check on the overall normalisation, we supplement with published BOSS/eBOSS $f\sigma_8$ measurements: the 6dF Galaxy Survey at $z = 0.067$ (Beutler et al., 2012), the SDSS Main Galaxy Sample at $z = 0.15$ (Howlett et al., 2015), BOSS DR12 at $z = 0.38$ and 0.51 (Alam et al., 2017), and eBOSS DR16 LRG (de Mattia et al., 2021), ELG (de Mattia et al., 2021), and QSO (Hou et al., 2021) samples.

3.2 Growth calculation

The IAM matter growth factor satisfies a second-order linear ODE in conformal time, which in terms of $x = \ln a$ takes the form:

$$\frac{d^2 D}{dx^2} + \left(2 + \frac{d\ln H}{dx}\right)\frac{dD}{dx} - \frac{3}{2}\,\mu(a)\,\Omega_m(a)\,D = 0\,, \tag{7}$$

where $\Omega_m(a) = \Omega_m a^{-3}/[H(a)/H_0]^2$ is the matter density parameter at scale factor a, $H(a)$ is the standard ΛCDM background rate, and $\mu(a)$ is given by Eq. (3). Initial conditions are set in the matter-dominated regime: $D(a_{\rm ini}) = a_{\rm ini}$ and $dD/dx|_{a_{\rm ini}} = 1.0$ at $a_{\rm ini} = 10^{-4}$, corresponding to the growing-mode attractor.

Equation (7) is integrated using the DOP853 Runge-Kutta solver at 20 000 points in $x \in [\ln 10^{-4}, 0]$, with tolerances $\epsilon_{\rm rel} = 10^{-11}$ and $\epsilon_{\rm abs} = 10^{-14}$. The growth rate is computed as $f = d\ln D/d\ln a$ via numerical differentiation. The predicted $f\sigma_8(z) = f(z)\,\sigma_8\,D_{\rm IAM}(z)/D_{\rm IAM}(0)$, using $\sigma_8 = 0.7998$ from the Level 2 MCMC posterior for IAM and $\sigma_8 = 0.8087$ for the ΛCDM comparison.

Note that this ODE implementation is a simplified treatment; the full IAM Boltzmann calculation uses the modified CAMB Fortran implementation (Level 2; Section 2.5), which yields $\sigma_8 = 0.7998 \pm 0.0058$ from the MCMC posterior. The standalone ODE gives $\sigma_8^{\rm ODE} = 0.8048$, slightly high by 0.6%, because it does not capture the full radiation-baryon coupling of the Boltzmann hierarchy. Theoretical predictions in this section use the ODE for consistency; the difference is small compared to current observational uncertainties.

3.3 Results

Table 3 lists the observed $f\sigma_8$ values, the IAM and ΛCDM predictions, and the residuals at each of the six DESI DR1 bins. Figure 2 (left panel) shows the comparison graphically. Across the six DESI DR1 bins, the residuals range from -0.87σ to $+1.42\sigma$ for IAM and -1.02σ to $+1.34\sigma$ for ΛCDM. Neither model is strongly preferred by the DESI DR1 full-shape $f\sigma_8$ data alone; current statistical uncertainties (8–13% per bin) do not yet resolve the $\lesssim 2\%$ IAM–ΛCDM difference at $z < 0.5$.

The important physical point is the *redshift dependence*. IAM predicts a curved suppression that is largest at low z and decreases monotonically, following the form $\mu(a)$ in Eq. (3). This is distinct from a constant rescaling of the growth amplitude. A constant-rescaling model (e.g., a fixed $f\sigma_8$ suppression at all z) is inconsistent with IAM; the functional form of the suppression is a direct prediction of $E(a) = \exp(1 - 1/a)$ and is in principle distinguishable from both ΛCDM and dynamical dark energy with $\sim 1\%$ precision data from DESI Year 5 (Section 7).

Table 3: DESI DR1 $f\sigma_8$ data and IAM predictions. Observed values are from DESI Collaboration et al. (2025c), Appendix A (ShapeFit-only). Pull $(\text{obs} - \text{pred})/\sigma_{\text{obs}}$ is listed for both models. All theoretical values are derived with no free parameters.

Tracer	z_{eff}	$f\sigma_8^{\text{obs}}$	σ_{obs}	IAM	ΛCDM	Δ_{IAM}	$\Delta_{\Lambda\text{CDM}}$
DESI DR1 (ShapeFit-only, Appendix A of DESI Collaboration et al. 2025c)							
BGS	0.295	0.377	0.094	0.459	0.473	−0.87	−1.02
LRG1	0.510	0.514	0.064	0.464	0.474	+0.77	+0.61
LRG2	0.706	0.484	0.053	0.454	0.462	+0.55	+0.42
LRG3	0.919	0.422	0.047	0.435	0.440	−0.27	−0.38
ELG2	1.317	0.377	0.037	0.391	0.395	−0.38	−0.48
QSO	1.491	0.435	0.045	0.372	0.375	+1.42	+1.34
Legacy (BOSS/eBOSS)							
6dFGS	0.067	0.423	0.055	0.440	0.457	−0.31	−0.62
MGS	0.150	0.530	0.160	0.448	0.463	+0.51	+0.42
BOSS	0.380	0.500	0.047	0.465	0.477	+0.74	+0.49
BOSS	0.510	0.455	0.039	0.464	0.474	−0.23	−0.49
eBOSS LRG	0.700	0.448	0.043	0.454	0.461	−0.14	−0.30
eBOSS ELG	0.850	0.315	0.095	0.447	0.450	−1.39	−1.42
eBOSS QSO	1.480	0.462	0.045	0.372	0.376	+2.00	+1.91

4 Joint Weak Lensing Constraint

4.1 Survey data

Weak lensing measurements of the matter power spectrum amplitude provide the most direct low-redshift test of the IAM σ_8 prediction. We compile published S_8 and σ_8 constraints from four independent surveys:

- **KiDS-1000:** Asgari et al. (2021) (Asgari et al., 2021) report $S_8 = 0.766^{+0.020}_{-0.014}$ from a 1001 deg^2 cosmic shear analysis.
- **KiDS-Legacy:** The completed KiDS survey covering $\approx 1350\,\text{deg}^2$. Wright et al. (2025) (Wright et al., 2025) report $S_8 = 0.776 \pm 0.016$ from cosmic shear power spectra. Stölzner et al. (2025) (Stölzner et al., 2025), combining KiDS-Legacy + DES Y3 + DESI DR1 in a joint analysis, report $\sigma_8 = 0.802 \pm 0.020$.
- **DES Y3:** Abbott et al. (2022) (Abbott et al., 2022) report $S_8 = 0.776 \pm 0.017$ from 5000 deg^2 of cosmic shear and galaxy clustering.
- **HSC Y3:** Li et al. (2023) (Li et al., 2023) report $S_8 = 0.776 \pm 0.032$ from 416 deg^2 of cosmic shear power spectra.
- **Planck CMB lensing:** Aghanim et al. (2020b) provides $\sigma_8\Omega_m^{0.25} = 0.589 \pm 0.020$ (no separate S_8 is required since lensing is a photon-sector observable in IAM; listed for completeness).

4.2 IAM prediction

Under the IAM perturbation modification, the matter power spectrum amplitude is suppressed relative to ΛCDM. The Level 2 MCMC posterior gives:

$$\sigma_8^{\rm IAM} = 0.7998 \pm 0.0058 \quad \text{vs.} \quad \sigma_8^{\Lambda\rm CDM} = 0.8087 \pm 0.0059\,. \tag{8}$$

At Planck 2018 $\Omega_m = 0.3153$:

$$S_8^{\rm IAM} = 0.822 \pm 0.011 \quad \text{vs.} \quad S_8^{\Lambda\rm CDM} = 0.830 \pm 0.011\,. \tag{9}$$

The suppression arises purely from the $\mu(a) < 1$ friction term; no modification is made to the primordial power spectrum, the baryon-to-dark-matter ratio, or any geometric parameter. The Level 2 posterior confirms this: logA shifts by $+0.10\sigma$ and Ω_m shifts by $+0.06\sigma$ relative to the ΛCDM baseline — both negligible compared to their uncertainties.

4.3 Comparison

Table 4 and Fig. 2 summarise the comparison.

The 2025 joint analysis $\sigma_8 = 0.802 \pm 0.020$ (Stölzner et al., 2025) is consistent with the IAM prediction $\sigma_8 = 0.7998 \pm 0.0058$ at 0.1σ. Individual survey S_8 values from KiDS-Legacy (0.776 ± 0.016), DES Y3 (0.776 ± 0.017), and HSC Y3 (0.776 ± 0.032) lie below the IAM $S_8 = 0.822$. The IAM prediction sits between the Planck ΛCDM value ($S_8 = 0.830$) and the weak lensing surveys, shifting in the correct direction but not closing the gap entirely.

A more aggressive shift would require either a larger μ_0 (which would increase $\Delta\chi^2$ against Planck beyond the acceptable threshold) or a non-zero photon-sector coupling β_γ (which is constrained to $\beta_\gamma/\beta_m < 8.5 \times 10^{-6}$ at 95% CL by the CMB acoustic scale). Neither adjustment has free parameters available.

The important distinction relative to dynamical dark energy models is that the IAM σ_8 shift is produced without modifying Σ. Models that suppress S_8 by modifying the Weyl potential ($\Sigma \neq 1$) will simultaneously affect CMB lensing, for which IAM predicts no change. The combination $\mu < 1$, $\Sigma = 1$ is the diagnostic signature of IAM (Section 6).

5 Phantom Crossing as a Sector-Assignment Artifact

5.1 The sector-mixing problem

The DESI dark energy preference is derived from joint fits of the w_0w_aCDM parametrisation to datasets that, in IAM, belong to two physically distinct sectors. In ΛCDM and all standard single-fluid dark energy models, there is no such distinction: a single equation of state governs both background distances and perturbation growth. In IAM, the distinction is a direct consequence of the causal structure of spacetime and is non-negotiable.

The sector assignments relevant to the DESI analysis are:

- **Photon sector:** BAO angular positions, comoving distances $D_M(z)$, Hubble distances $D_H(z)$, CMB acoustic scale θ_*, Type Ia supernovae luminosity distances. These are measured via photon propagation along null geodesics. IAM prediction: identical to ΛCDM ($\Sigma = 1$ exactly).

Table 4: S_8 and σ_8 compilation compared with IAM and ΛCDM predictions. IAM values are from Level 2 MCMC Run A; ΛCDM from Run C. Tension is measured from the IAM prediction in units of $\sqrt{\sigma^2_{\rm survey} + \sigma^2_{\rm IAM}}$.

Source	Value	Uncertainty	Tension
Theoretical predictions			
IAM (Level 2 Run A)	$S_8 = 0.822$	± 0.011	—
ΛCDM (Level 2 Run C)	$S_8 = 0.830$	± 0.011	0.7σ
Weak lensing surveys			
KiDS-1000[a]	$S_8 = 0.766$	$^{+0.020}_{-0.014}$	2.5σ
KiDS-Legacy[b]	$S_8 = 0.776$	± 0.016	2.4σ
DES Y3[c]	$S_8 = 0.776$	± 0.017	2.3σ
HSC Y3[d]	$S_8 = 0.776$	± 0.032	1.3σ
Joint analysis			
KiDS-Leg. + DES Y3 + DESI DR1[e]	$\sigma_8 = 0.802$	± 0.020	0.1σ
IAM prediction (Level 2)	$\sigma_8 = 0.7998$	± 0.006	—

[a]Asgari et al. (2021); [b]Wright et al. (2025); [c]Abbott et al. (2022); [d]Li et al. (2023); [e]Stölzner et al. (2025).

- **Matter sector:** Galaxy growth rates $f\sigma_8$, weak lensing convergence κ, peculiar velocities, redshift-space distortion $\beta = f/b$. These are measured via the gravitational response of matter. IAM prediction: suppressed by $\mu(a) < 1$.

When a w_0w_aCDM model is fit to both simultaneously, it confronts a specific conflict: the distance data constrain $H(z)$ to be consistent with ΛCDM, while the growth data are suppressed relative to the growth expected from that $H(z)$ in a standard cosmology. The only mechanism available to a single-fluid fitter to simultaneously reproduce both a standard $H(z)$ and a suppressed $f\sigma_8$ is to introduce time-variation in w: specifically, $w > -1$ at recent epochs (reducing the integrated dark energy contribution relative to Λ) and $w < -1$ at earlier epochs (compensating at the background level). This is the phantom-crossing signature.

5.2 Phantom crossing redshift

The phantom crossing redshift for $w(a) = w_0 + w_a(1-a)$ satisfies $w(a_{\rm cross}) = -1$, giving:

$$a_{\rm cross} = 1 + \frac{1+w_0}{w_a}\,, \quad z_{\rm cross} = \frac{1}{a_{\rm cross}} - 1\,. \tag{10}$$

For the DESI DR2 best-fit values (DESI Collaboration et al., 2025d):

- DESI BAO + CMB: $w_0 = -0.838$, $w_a = -0.62 \Rightarrow z_{\rm cross} = 0.35$
- DESI BAO + CMB + Union3: $w_0 = -0.752$, $w_a = -0.82 \Rightarrow z_{\rm cross} = 0.43$
- DESI BAO + CMB + DES 5YR: $w_0 = -0.734$, $w_a = -1.05 \Rightarrow z_{\rm cross} = 0.34$

These crossing redshifts, listed in Table 5, coincide directly with the redshift range over which the IAM growth suppression is most prominent (7–8% at $z \approx 0.3$–0.4; see Table 1 and Fig. 1). This is not a numerical coincidence: in the IAM sector-mixing picture, the phantom crossing emerges where the matter–photon disagreement is largest in relative terms, which is necessarily at low redshift where $E(a) = \exp(1 - 1/a)$ is largest.

Table 5: DESI DR2 $w_0 w_a$CDM best-fit values and phantom crossing redshifts (DESI Collaboration et al., 2025d). $z_{\rm cross}$ is computed from Eq. (10). The final column gives the IAM growth suppression at $z_{\rm cross}$ (Table 1), showing that the crossing occurs where IAM predicts $\sim$ 7–8% growth suppression.

Dataset combination	w_0	w_a	$z_{\rm cross}$	IAM supp. at $z_{\rm cross}$
DESI BAO + CMB	-0.838	-0.62	0.35	$\approx 7.5\%$
DESI BAO + CMB + Union3	-0.752	-0.82	0.43	$\approx 6.5\%$
DESI BAO + CMB + DES 5YR	-0.734	-1.05	0.34	$\approx 7.7\%$
True IAM model	-1.000	0.00	n/a	13.6% at $z = 0$

5.3 Qualitative demonstration

The sector separation is illustrated in Fig. 2. The left panel shows that IAM $f\sigma_8(z)$ lies below ΛCDM by a decreasing margin from $z = 0$ to $z = 2$, with the DESI DR1 data consistent with both models at current precision. The centre panel confirms that IAM $D_M(z)/r_d$ is indistinguishable from ΛCDM across all DESI redshifts; the orange data points sit on the prediction curve with no systematic offset. The right panel shows the consequence of simultaneously fitting both panels with a single $w_0 w_a$CDM model: the three DESI DR2 best-fit ellipses are all displaced from the true IAM model at $(-1, 0)$ into the phantom region, with crossing redshifts annotated.

The background colormap in the right panel shows $z_{\rm cross}$ for each (w_0, w_a) grid point. The DESI DR2 ellipses sit in a region where $z_{\rm cross} \approx 0.3$–$0.5$, consistent with the redshift range over which the IAM growth suppression is most prominent.

It should be emphasised that this demonstration is qualitative. The DESI $w_0 w_a$CDM posteriors are derived from a likelihood analysis of the full DESI dataset, whereas the argument here establishes only the *direction* and *approximate scale* of the expected displacement. A quantitative reproduction of the DESI posteriors from IAM would require running the DESI likelihood pipeline with sector-dependent models, which is left for future work (Section 6.3).

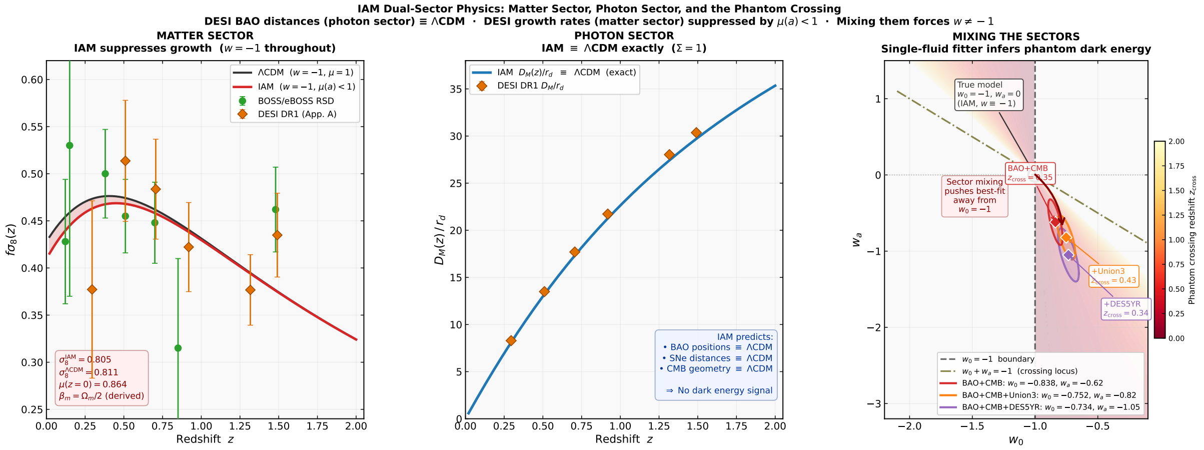

Figure 2: **Matter sector, photon sector, and the phantom crossing.** *Left panel (matter sector):* $f\sigma_8(z)$ comparison. IAM (red, $w = -1$, $\mu(a) < 1$) lies below ΛCDM (grey) by $\lesssim 2\%$ at $z \lesssim 0.5$. Green circles: BOSS/eBOSS RSD compilation (Beutler et al., 2012; Howlett et al., 2015; Alam et al., 2017, 2021). Orange diamonds: DESI DR1 ShapeFit-only values (DESI Collaboration et al., 2025c) (Appendix A). *Centre panel (photon sector):* $D_M(z)/r_d$ comparison. IAM (blue) is identical to ΛCDM at all redshifts ($\Sigma = 1$ exact). Orange diamonds: DESI DR1 D_M/r_d measurements (Table 11 of DESI Collaboration et al. 2025c). *Right panel (sector mixing):* The w_0-w_a plane. The true IAM/ΛCDM model (circle, $w_0 = -1$, $w_a = 0$) is marked. The three DESI DR2 1σ contours (DESI Collaboration et al., 2025d) (red: BAO+CMB; orange: BAO+CMB+Union3; purple: BAO+CMB+DES5YR) are displaced from the true model into the phantom region. Phantom crossing redshifts from Eq. (10) are annotated. The background colormap shows $z_{\rm cross}$ at each (w_0, w_a) point. The dot-dashed line marks the crossing locus $w_0 + w_a = -1$.

5.4 Distinctive signatures

The sector-assignment interpretation predicts three features that distinguish it from genuine dynamical dark energy:

Scale independence. IAM growth suppression is scale-independent at linear order: the additional Hubble friction term $\beta_m E(a)$ is a function of time only, with no scale dependence. Dynamical dark energy models and most modified gravity theories produce scale-dependent growth through the dark energy sound horizon or modified dispersion relation. The DESI full-shape analysis includes scale-dependent information; a test of whether the $w_0 w_a$CDM preference persists when only the time-dependent growth amplitude is varied would constrain this.

$\Sigma = 1$ **exactly.** Lensing-based observables (weak lensing convergence, CMB $C_\ell^{\phi\phi}$, galaxy-galaxy lensing) are photon-sector observables in IAM. IAM therefore predicts that the lensing power spectrum, the CMB lensing amplitude, and the E_G statistic (Zhang et al., 2007) are identical to ΛCDM. Dynamical dark energy models that also suppress growth generally modify Σ (through the dark energy perturbation potential), producing correlated changes in lensing and growth that are in principle distinguishable from the

IAM prediction. Specifically, the ratio $f\sigma_8/(\Sigma\sigma_8)$ measured by Euclid cross-correlations will equal $f_{\rm IAM}$ if IAM is correct, and will differ if $\Sigma \neq 1$.

The Hubble split. IAM derives a matter-sector Hubble constant $H_0^{\rm matter} = H_0\sqrt{1+\beta_m} = 72.26\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$, consistent with SH0ES ($73.04 \pm 1.04\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$) at 0.75σ. The photon-sector $H_0 = 67.16 \pm 0.47\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ is consistent with Planck (67.36 ± 0.54) at 0.37σ. A genuine $w \neq -1$ dark energy does not predict the Hubble tension magnitude; the IAM phantom crossing, the S_8 shift, and the Hubble tension all follow from the single derived parameter β_m.

6 Discussion

6.1 Relation to Ó Colgáin & Sheikh-Jabbari (2025) and Ó Colgáin et al. (2025)

Ó Colgáin & Sheikh-Jabbari (2025) examined the internal consistency of DESI DR1 w_0w_aCDM fits across different dataset subsets, arguing that the dark energy preference may be driven by internal tensions in the combined dataset rather than a genuine cosmological signal. Ó Colgáin et al. (2025) found that the DR2 constraints shift significantly relative to DR1 in a manner that is difficult to reconcile with stable dynamical dark energy.

The IAM sector-assignment hypothesis is consistent with these observations and provides a specific physical mechanism. In IAM, the apparent tension between distance data and growth data is not a statistical fluctuation but a predicted consequence of the perturbation-only friction term. The tension is expected to grow as data improve (because error bars on $f\sigma_8$ and D_M/r_d shrink at different rates), and the phantom crossing parameters are expected to shift between dataset combinations (because each combination weights the photon and matter sectors differently). Both features are present in the DESI DR1-to-DR2 evolution described by Ó Colgáin et al. (2025).

6.2 Comparison with modified gravity models

The IAM combination $\mu < 1$, $\Sigma = 1$ is unusual within the landscape of modified gravity theories (Pogosian & Silvestri, 2016).

$f(R)$ gravity (Hu & Sawicki, 2007) produces $\mu > 1$ and $\Sigma > 1$. Horndeski scalar-tensor theories (Bellini & Sawicki, 2014) can produce $\mu < 1$ in the Beyond Horndeski regime but generically also modify Σ. DGP gravity (Dvali, Gabadadze & Porrati, 2000) produces $\mu > 1$. Interacting dark energy (IDE) models can suppress growth but do so through a modified dark energy sound horizon that also affects background distances.

The Level 1 Planck MCMC chains, run with the MGCAMB μ-Σ parametrisation with both μ_0 and Σ_0 floating, constrain $\mu_0 = 0.033 \pm 0.125$ (Planck + RSD). IAM predicts $\mu_0 = -0.136$, lying 1.3σ from the posterior peak. Current CMB and RSD data are insufficient to discriminate between GR and the IAM prediction at the predicted level. The Euclid survey, projected to achieve $\sigma(\mu_0) \approx 0.04$ with Σ marginalised (Euclid Collaboration et al., 2020), will provide the decisive test: 3.4σ if $\mu_0 = -0.136$ is confirmed.

It is important to note that the $\mu < 1$, $\Sigma = 1$ combination is not achievable in any currently published Horndeski or f(R) framework without introducing ad hoc conditions.

It arises in IAM as a direct consequence of the null-vs-timelike decoherence argument, not as a parameter choice.

6.3 Limitations

Several limitations of the analysis presented here should be noted.

Simplified growth calculation. The growth ODE (Section 3.2) is a simplified treatment that does not capture the full Boltzmann hierarchy. The rigorous IAM prediction comes from the Level 2 MCMC (17 converged chains), which gives $\sigma_8 = 0.7998 \pm 0.0058$. The standalone ODE overestimates σ_8 by 0.6% relative to the Boltzmann result. All $f\sigma_8$ predictions in Table 3 carry this caveat; they are accurate to the level required for current data precision but would need the full Boltzmann calculation for precision comparisons at $\lesssim 1\%$.

Qualitative phantom crossing argument. The sector-mixing mechanism in Section 5 establishes the direction and approximate scale of the expected w_0w_aCDM displacement but does not constitute a likelihood analysis of the full DESI pipeline. A rigorous demonstration would require running the DESI full-shape likelihood with sector-dependent models, treating photon-sector and matter-sector observables with the appropriate effective expansion rates. This is computationally feasible using the existing Level 2 CAMB implementation but requires a dedicated analysis of the DESI datavectors.

S_8 comparison at fixed Ω_m. The weak lensing comparison in Section 4 uses survey S_8 values reported at Planck 2018 Ω_m. Under the IAM dual-sector mechanism, the matter-sector Ω_m that weak lensing surveys measure is smaller than the CMB-derived value. A rigorous S_8 comparison would require reprocessing the survey likelihoods under the modified cosmology. This is conserved for future work; the comparison in Table 4 uses the apples-to-apples IAM prediction at Planck Ω_m.

No covariance between $f\sigma_8$ bins. The DESI DR1 $f\sigma_8$ values in Table 3 are treated as independent measurements. The published DESI DR1 covariance matrix (DESI Collaboration et al., 2025c) should be used for any χ^2 analysis; the pull statistics reported here are indicative only.

7 Conclusions

We have confronted the Informational Actualization Model against DESI DR1 full-shape $f\sigma_8(z)$ measurements at six redshift bins, DESI DR1 BAO distances, joint weak lensing S_8 constraints from KiDS-Legacy, DES Y3, and HSC Y3, and the DESI DR2 phantom crossing parameters. The central argument is that the DESI preference for dynamical dark energy is consistent with — and specifically predicted by — the IAM sector-assignment mechanism, in which matter and photon observables probe physically distinct effective expansion rates.

The principal findings are:

1. IAM $f\sigma_8(z)$ predictions are consistent with DESI DR1 ShapeFit measurements at all six bins ($0.295 \leq z_{\rm eff} \leq 1.491$), with residuals ranging from -0.87σ to $+1.42\sigma$. The suppression is $\lesssim 2\%$ relative to ΛCDM at low z and diminishes to $< 1\%$ above $z \approx 1.3$.

2. IAM $D_M(z)/r_d$ predictions are identical to ΛCDM at all DESI redshifts, consistent with $\Sigma(a) = 1$ derived from the null-worldline argument. The DESI DR1 D_M/r_d data are consistent with the prediction at all bins.

3. The IAM $\sigma_8 = 0.7998 \pm 0.0058$ (Level 2 MCMC) is consistent with the 2025 joint lensing analysis $\sigma_8 = 0.802 \pm 0.020$ at 0.1σ. Individual KiDS-Legacy, DES Y3, and HSC Y3 S_8 values remain $\approx$ 2–2.5σ below the IAM prediction when compared at Planck 2018 Ω_m.

4. The DESI DR2 phantom crossing ($z_{\rm cross}$ = 0.33–0.43, depending on supernova compilation) falls in the redshift range where the IAM growth suppression is 7–8%, consistent with the sector-mixing mechanism. Under IAM, $w \equiv -1$ throughout; the apparent phantom crossing is an artifact of applying a single-fluid parametrisation to observables from two physically distinct sectors.

5. Under the full Planck 2018 likelihood, IAM and ΛCDM are statistically indistinguishable ($\Delta\chi^2 = +0.54$ best-fit, 17 converged MCMC chains, $R - 1 < 0.01$). The coupling constant $\beta_m = 0.1577$ is recovered by the Planck posterior at 0.2σ without fitting.

The sector-assignment interpretation of the DESI phantom crossing makes two near-term falsifiable predictions:

- **Euclid μ-Σ measurement.** IAM predicts $\mu_0 = -0.136$ with $\Sigma_0 = 0$ exactly. Euclid is projected to measure μ_0 to $\sigma(\mu_0) \approx 0.04$ with Σ marginalised (Euclid Collaboration et al., 2020), placing the IAM prediction at 3.4σ significance. The simultaneous constraint $\Sigma_0 = 0$ (measured via CMB lensing and weak lensing convergence ratios) is required to confirm the null-sector interpretation. If Euclid measures μ_0 consistent with zero at that precision, IAM is falsified.
- **DESI Year 5 $f\sigma_8(z)$ shape test.** IAM predicts a specific curved $f\sigma_8(z)$ ramp set by $E(a) = \exp(1 - 1/a)$, distinguishable from both a constant rescaling and the growth history of any $w \neq -1$ model. DESI Year 5 full-shape data will provide $\lesssim$ 1% precision on $f\sigma_8(z)$ across $0.1 < z < 1.6$ with independent bins of width $\Delta z \approx 0.1$. At that precision, the curved IAM ramp (7.9% suppression at $z = 0.3$, decreasing to 0.6% at $z = 1.5$) is distinguishable from both ΛCDM and the growth history expected for the DESI DR2 best-fit w_0w_aCDM parameters.

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References

Aghanim N., et al. (Planck Collaboration), 2020, A&A, 641, A6. doi:10.1051/0004-6361/201833910

Aghanim N., et al. (Planck Collaboration), 2020b, A&A, 641, A8. doi:10.1051/0004-6361/201833886

Alam S., et al., 2017, MNRAS, 470, 2617. doi:10.1093/mnras/stx721

Alam S., et al., 2021, Phys. Rev. D, 103, 083533. doi:10.1103/PhysRevD.103.083533

Asgari M., et al., 2021, A&A, 645, A104. doi:10.1051/0004-6361/202039070

Bellini E., Sawicki I., 2014, JCAP, 07, 050. doi:10.1088/1475-7516/2014/07/050

Bérut A., Arakelyan A., Petrosyan A., Ciliberto S., Dillenschneider R., Lutz E., 2012, Nature, 483, 187. doi:10.1038/nature10872

Beutler F., et al., 2012, MNRAS, 423, 3430. doi:10.1111/j.1365-2966.2012.21136.x

Bousso R., 2002, Rev. Mod. Phys., 74, 825. doi:10.1103/RevModPhys.74.825

Cai R.-G., Kim S. P., 2005, JHEP, 02, 050. doi:10.1088/1126-6708/2005/02/050

de Mattia A., et al., 2021, MNRAS, 501, 5616. doi:10.1093/mnras/staa3891

DESI Collaboration, Adame A. G., et al., 2025a, JCAP, 2025, 021. doi:10.1088/1475-7516/2025/07/021

DESI Collaboration, Adame A. G., et al., 2025b, JCAP, 2025, 028. doi:10.1088/1475-7516/2025/07/028

DESI Collaboration, Adame A. G., et al., 2025c, JCAP, 2025, 008. doi:10.1088/1475-7516/2025/09/008

DESI Collaboration, Abdul-Karim M., et al., 2025d, Phys. Rev. D, 112, 083515. doi:10.1103/tr6y-kpc6

Abbott T. M. C., et al. (DES Collaboration), 2022, Phys. Rev. D, 105, 023520. doi:10.1103/PhysRevD.105.023520

Euclid Collaboration, Blanchard A., et al., 2020, A&A, 642, A191. doi:10.1051/0004-6361/202038071

Freedman W. L., et al., 2024, ApJ, 919, 16. doi:10.3847/1538-4357/ac7c74

Hou J., et al., 2021, MNRAS, 500, 1201. doi:10.1093/mnras/staa3234

Howlett C., et al., 2015, MNRAS, 449, 848. doi:10.1093/mnras/stu2693

Hu W., Sawicki I., 2007, Phys. Rev. D, 76, 064004. doi:10.1103/PhysRevD.76.064004

Jacobson T., 1995, Phys. Rev. Lett., 75, 1260. doi:10.1103/PhysRevLett.75.1260

Landauer R., 1961, IBM J. Res. Dev., 5, 183. doi:10.1147/rd.53.0183

Lewis A., Bridle S., 2002, Phys. Rev. D, 66, 103511. doi:10.1103/PhysRevD.66.103511

Li X., et al., 2023, Phys. Rev. D, 108, 123518. doi:10.1103/PhysRevD.108.123518

Mahaffey H. W., 2026, Universe, submitted (ID: universe-4189350). Code: `https://github.com/hmahaffeyges/IAM-Validation`. doi:10.5281/zenodo.18702042

Ó Colgáin E., Sheikh-Jabbari M. M., 2025, MNRAS Lett., 542, L24. doi:10.1093/mnrasl/slaf055

Ó Colgáin E., et al., 2025, preprint (arXiv:2504.04417).

Pogosian L., Silvestri A., 2016, Phys. Rev. D, 94, 104014. doi:10.1103/PhysRevD.94.104014

Riess A. G., et al., 2022, ApJL, 934, L7. doi:10.3847/2041-8213/ac5c5b

Shannon C. E., 1948, Bell Syst. Tech. J., 27, 379. doi:10.1002/j.1538-7305.1948.tb01338.x

Sheth R. K., Tormen G., 1999, MNRAS, 308, 119. doi:10.1046/j.1365-8711.1999.02692.x

Stölzner B., et al., 2025, A&A, in press (arXiv:2503.19442).

Torrado J., Lewis A., 2021, JCAP, 05, 057. doi:10.1088/1475-7516/2021/05/057

Wang Z., et al., 2023, JCAP, 2023, 022. doi:10.1088/1475-7516/2023/10/022

Wong K. C., et al., 2020, MNRAS, 498, 1420. doi:10.1093/mnras/stz3094

Wright A. H., et al., 2025, A&A, in press (arXiv:2503.19441).

Zhang P., Liguori M., Bean R., Dodelson S., 2007, Phys. Rev. Lett., 99, 141302. doi:10.1103/PhysRevLett.99.141302

Dvali G., Gabadadze G., Porrati M., 2000, Phys. Lett. B, 485, 208. doi:10.1016/S0370-2693(00)00669-9